HW 4

Tuesday, March 3, 2026 4:24 PM

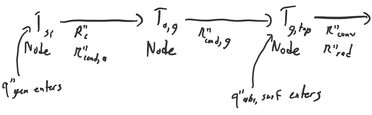
1) (a) Assumptions:

i Steady-state

ii One-dimensional heat transfer

iii Constant properties (k, h, etc)

iv Uniform heat generation (silicon thin enough to be treated as a plane



Calculate all thermal resistances ($m^2 \cdot K/W$)

Contact Resistance: $R_c'' = 10^{-4} m^2 \cdot K/W$

Adhesive Conduction: $R''_{cond,a} = \frac{L_a}{k_a} = \frac{.1 \cdot 10^{-3}}{1 W/m \cdot K} = 10^{-4} m^2 K/W$

Glass Conduction: $R''_{cond,g} = L_g/k_g = 3 \times 10^{-3} / 1.5 W/m \cdot K = 2 \times 10^{-3} m^2 \cdot K/W$

Surface Convection: $R''_{conv} = 1/h = .1 m^2 \cdot K/W$

Surface Radiation: $R''_{rad} = 1/h_{rad} = .2 m^2 \cdot K/W$

Equivalent Surface Resistance: $R''_{sur} = \frac{1}{R''_{at}} = \frac{1}{R''_{conv}} + \frac{1}{R''_{rad}} = 10 + 5 = 15$

$\rightarrow R''_{sur} = .0667 m^2 \cdot K/W$

(b) Calculate heat fluxes

Total absorbed in Silicon: $q''_{abs,si} = .85G = .85(800) = 680 W/m^2$

Absorbed at glass surface: $q''_{abs,surf} = .1(G) = 80 W/m^2$

$q''_{th} = q''_{abs,si} - P_{elec} = q''_{abs,si}(1-\eta)$

$q''_{tot} = q''_{abs,surf} + q''_{abs,si}(1-\eta) = \frac{T_{g,top} - T_\infty}{R''_{sur}}$

$T_{si} = T_{g,top} + (q''_{abs,si}(1-\eta)) \cdot (R''_{cond,g} + R''_{cond,a} + R''_c)$

Use $\eta = .553 - .001 T_{si}$

$T_{si} = T_{g,top} + (q''_{abs,si}(1 - .553 + .001 T_{si})) \cdot (R''_{cond,g} + R''_{cond,a} + R''_c)$

Used a sys. of eqns. in calculator to solve this:

$T_{g,top} = 330.3 K$

$T_{si} = 331.4 K$

(c) Use $T_{si} = 331.4 K$

$\eta = .553 - .001(331.4) = .2216$ (22.16%)

Electricity Power Flux: $P''_{elec} = \eta \cdot .85G = 150.7 W/m^2$

2) $D = .01 m$

Thermal cond: $80 W/m^2 K$ (k)

Insul. thickness (L_{ins}): $200 mm = .2 m$

$T_{wall} = 200°C$

$T_\infty = 20°C$

$h = 20 W/m^2 K$

Target $T_0 \leq 100°C$

(a) Must balance heat flow

$q_{cond} = \frac{kA_c(T_w - T_0)}{L_{ins}}$ $A_c = \frac{\pi D^2}{4}$

$m = \sqrt{\frac{hP}{kA_c}} = \sqrt{\frac{4h}{kD}}$

$q_{fin} = M \tanh(mL_0)$

$M = \sqrt{h_p k A_c} \theta_0 = \sqrt{h(\pi D) k \left(\frac{\pi D^2}{4}\right)} (T_0 - T_\infty)$

Set $q_{cond} = q_f$ and solve

$\frac{kA_c(T_w - T_0)}{L_{ins}} = \sqrt{h_p k A_c} \tanh(mL_0)(T_0 - T_\infty)$

$\frac{T_w - T_0}{L_{ins}} = m \tanh(mL_0)(T_0 - T_\infty)$

$T_0 = \frac{T_w + T_\infty [m L_{ins} \tanh(mL_0)]}{1 + m L_{ins} \tanh(mL_0)}$

(b) First: $m = \sqrt{\frac{4 \cdot 20}{80 \cdot .01}} = \sqrt{\frac{80}{.8}} = 10 m^{-1}$

$mL_0 = 2$

$mL_{ins} = 2$

$\tanh(2) = .964$

$T_0 = \frac{200 + 20(2 \cdot .964)}{1 + 2 \cdot .964} = 81.5°C$

(c) Set $T_0 = 100°C$

$\frac{200 - 100}{-2} = 10 \tanh(10L_0)(100 - 20)$

$500 = 800 \tanh(10L_0)$

$L_0 = 73.3 mm$

3) Given

$k = 400 W/m \cdot K$

$s = 10 mm = .1 m$

$h = 100 W/m^2 K$

$\eta_f = .6$

Convection tip

(a) $A_c = s^2 = .0001$ $P = 4s = .04 m$

$m = \sqrt{\frac{hP}{kA_c}} = \sqrt{\frac{4}{.04}} = 10 m^{-1}$

$L_c = L + \frac{A_c}{P} = L + \frac{s^2}{4s} = L + \frac{s}{4} = L + .0025$

$\eta_f = \frac{\tanh(mL_c)}{mL_c}$

$\eta_f = .6 \rightarrow L_c = .1515$

$L = L_c - .0025 = \boxed{.149 m}$

(b) $R_f = \frac{1}{\eta_f h A_f}$

$A_f = P \cdot L + A_c = (.04 \cdot .149) + .0001 = .005 96 + .0001 = .00606$

$R_f = \frac{1}{.6 \cdot 100 \cdot .00606} = \frac{1}{.3636} = 2.75 K/W$

$\epsilon_f = \frac{R_{f,base}}{R_f} = \frac{1/(hA_c)}{R_f} = \frac{\eta_f A_f}{A_c} = 36.36$